

L'HOPITAL'S RULE & INDETERMINATE FORMS

In computing limits of the form

$$\lim_{x \rightarrow 0} \frac{f(x)}{g(x)}, \quad \lim_{x \rightarrow \infty} \frac{f(x)}{g(x)}, \quad \lim_{x \rightarrow x_0} \frac{f(x)}{g(x)}$$

we encounter output

of the form $\frac{0}{0}$, $\frac{\infty}{\infty}$, 0^0 , 0^∞ etc called indeterminate forms.

LIMITS	INDETERMINATE FORM
(1) $\lim_{x \rightarrow 0} \frac{\sin x}{x}$	$\neq \frac{0}{0}$
(2) $\lim_{x \rightarrow \infty} \frac{2x^2 - 2x + 3}{x^2 + 4x + 4}$	$\neq \frac{\infty}{\infty}$
(3) $\lim_{x \rightarrow 0^+} x \ln x$	$\neq 0 \cdot \infty$
(4) $\lim_{x \rightarrow 0^+} (\csc x - \cot x)$	$\neq \infty - \infty$
(5) $\lim_{x \rightarrow 0^+} x^{ax}$	$\neq 0^0$
(6) $\lim_{x \rightarrow \infty} x^{\frac{1}{x}}$	$\neq \infty^0$
(7) $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$	$\neq 1^\infty$

In the case of indeterminate cases $\frac{0}{0}$ and $\frac{\infty}{\infty}$ we apply L'Hopital's rule, which is stated as a theorem below:

THEOREM

Let $f(x)$ and $g(x)$ are continuous functions on $x \in \mathbb{R}$ and differentiable at x_0 , if $\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)}$ is indeterminate $\frac{0}{0}$ or $\frac{\infty}{\infty}$ satisfying

$$\lim_{x \rightarrow x_0} \frac{f^{(n)}(x)}{g^{(n)}(x)} = L, \quad L \in \mathbb{R}, n \in \mathbb{N}$$

Example 1

Calculate the following limits:

$$(a) \lim_{x \rightarrow 1} \frac{x^3 - 5x^2 + 6x - 2}{x^5 - 4x^4 + 7x^2 - 9x + 5}$$

$$(b) \lim_{x \rightarrow 0} \frac{\sin x}{x}$$

$$(c) \lim_{x \rightarrow 0} \frac{\cos x - 1}{x}$$

$$(d) \lim_{x \rightarrow 0^+} \frac{\sqrt{x}}{\sin 3\sqrt{x}}$$

$$(e) \lim_{x \rightarrow 0} \frac{\sin x}{x^3}$$

$$(f) \lim_{x \rightarrow 0} \frac{\sin 3x - 3x}{x^3}$$

Solution

$$(a) \lim_{x \rightarrow 1} \frac{x^3 - 5x^2 + 6x - 2}{x^5 - 4x^4 + 7x^2 - 9x + 5} \neq \frac{0}{0} \text{ indeterminate form}$$

Apply L'Hopital rule:

$$\lim_{x \rightarrow 1} \frac{3x^2 - 10x + 6}{5x^4 - 16x^3 + 14x - 9} = \frac{1}{6}$$

$$(b) \lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{\cos x}{1} = \frac{1}{1} = 1$$

L'Hopital

$$(c) \lim_{x \rightarrow 0} \frac{\cos x - 1}{x} = \lim_{x \rightarrow 0} \frac{\sin x}{1} = \frac{0}{1} = 0$$

$$(d) \lim_{x \rightarrow 0^+} \frac{\sqrt{x}}{\sin 3\sqrt{x}} \neq \frac{0}{0} \text{ indeterminate form}$$

Apply L'Hopital rule

$$= \lim_{x \rightarrow 0^+} \frac{\frac{1}{2\sqrt{x}}}{\frac{1}{3} \cdot 3 \cos 3\sqrt{x}}$$

$$= \lim_{x \rightarrow 0^+} \frac{1}{3 \cos 3\sqrt{x}} = \frac{1}{3}$$

$$(e) \lim_{x \rightarrow 0} \frac{\sin x}{x^3} \neq \frac{0}{0} \text{ indeterminate form}$$

$$= \lim_{x \rightarrow 0} \frac{\cos x}{3x^2} = \infty$$

$$(f) \lim_{x \rightarrow 0} \frac{\sin 3x - 3x}{x^3} \neq \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{3 \cos 3x - 3}{3x^2} \neq \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{-9 \sin 3x}{6x} \neq \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{-27 \cos 3x}{6} = \frac{-27}{6} = -\frac{9}{2}$$

Note: We can apply L'Hopital rule more than once.

Example 2

Calculate the following limits:

(a) $\lim_{x \rightarrow 1} \frac{\ln x}{x-1}$

(b) $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$

Solution

(a) $\lim_{x \rightarrow 1} \frac{\ln x}{x-1}$ ($\neq \frac{0}{0}$) indeterminate form

$$\lim_{x \rightarrow 1} \frac{\frac{1}{x}}{1} = \frac{1}{1} = 1$$

Apply L'Hopital rule

(b) $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x$ ($\neq 1^\infty$) indeterminate form

Let $y = \left(1 + \frac{1}{x}\right)^x \Rightarrow \ln y = x \ln \left(1 + \frac{1}{x}\right) = \frac{\ln \left(1 + \frac{1}{x}\right)}{\frac{1}{x}}$

$$\lim_{x \rightarrow \infty} \ln y = \lim_{x \rightarrow \infty} \frac{\ln \left(1 + \frac{1}{x}\right)}{\frac{1}{x}}$$
 ($\neq \frac{-\infty}{0}$) indeterminate form

$$= \lim_{x \rightarrow \infty} \frac{-\frac{1}{x^2}}{-\frac{1}{x^2}}$$

Apply L'Hopital's rule.

$$= \lim_{x \rightarrow \infty} \frac{1}{\left(1 + \frac{1}{x}\right)} = \frac{1}{1} = 1$$

Since $\lim_{x \rightarrow \infty} \ln y = \lim_{x \rightarrow \infty} \ln \left(1 + \frac{1}{x}\right)^x = 1$

then $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e^1$

Example 3

Calculate the following limits:

(a) $\lim_{x \rightarrow \infty} \frac{e^x}{x}$

(b) $\lim_{x \rightarrow 0^+} x \ln x$

(c) $\lim_{x \rightarrow 0^+} x^{ax} \quad (a \neq 0)$

(d) $\lim_{x \rightarrow \infty} \frac{3x}{5x+2\sin x}$

(e) $\lim_{x \rightarrow 0^+} (\csc x - \cot x)$

(f) $\lim_{x \rightarrow \infty} x^a e^{-bx}$

Solution

(a) $\lim_{x \rightarrow \infty} \frac{e^x}{x} \quad (\neq \frac{\infty}{\infty})$ indeterminate form
 $\searrow$ Apply L'Hopital rule:
 $= \lim_{x \rightarrow \infty} \frac{e^x}{1} = \infty$

(b) $\lim_{x \rightarrow 0^+} x \ln x$
 $= \lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{x}} \quad (\neq \frac{-\infty}{\infty})$ indeterminate form
 $= \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-\frac{1}{x^2}}$

(c) $\lim_{x \rightarrow 0^+} x^{ax} \quad (a \neq 0)$
 $\neq 0^0$ (indeterminate form)

$= \lim_{x \rightarrow 0^+} (-x) = 0$

Let $y = x^{ax} \Rightarrow \ln y = ax \ln x = a \frac{\ln x}{\frac{1}{x}}$

$\lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} \frac{a \ln x}{\frac{1}{x}} \quad (\neq \frac{-\infty}{\infty})$ indeterminate form
 $\searrow$ Apply L'Hopital rule:
 $= \lim_{x \rightarrow 0^+} \frac{a(\frac{1}{x})}{-\frac{1}{x^2}}$

$= \lim_{x \rightarrow 0^+} (-ax) = 0$

Since $\lim_{x \rightarrow 0^+} \ln[x^{ax}] = 0$

Hence $\lim_{x \rightarrow 0^+} x^{ax} = e^0 = 1$

(e) $\lim_{x \rightarrow 0^+} (\csc x - \cot x)$
 $= \lim_{x \rightarrow 0^+} \left(\frac{1}{\sin x} - \frac{\cos x}{\sin x} \right)$

$= \lim_{x \rightarrow 0^+} \frac{1 - \cos x}{\sin x} \quad (\neq \frac{0}{0})$

$= \lim_{x \rightarrow 0^+} \frac{\sin x}{\cos x} = \frac{0}{1} = 0$

(d) $\lim_{x \rightarrow \infty} \frac{3x}{5x+2\sin x} = \lim_{x \rightarrow \infty} \frac{3}{5+2(\frac{\sin x}{x})} = \frac{3}{5}$

"L'Hopital rule does not apply"